MILITARY SPECIFICATION

BUILDING, RELOCATABLE, PRE-FABRICATED, READY CUT, ALUMINUM, RIGID FRAME GENERAL PURPOSE

1. SCOPE

1.1 This specification covers one type of a prefabricated, unassembled, rigid frame, aluminum building having vertical side walls, vertical end walls and gable roof. This is a general specification and intended for use with a detailed specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

FF-H-106	Hardware, Builders Locks and Door Trim
FF-H-116	Hinges, Hardware Builders
HH-I-521	Insulation, Building, Mineral Wool, Batts, Loose-Fill and Granular Fill
HH-I-558	Insulation, Blocks, Boards, Blankets, Felts, Sleeving (Pipe and Tube Covering) Thermal (Mineral Fiber, Industrial Type)
QQ-A-200	Aluminum Alloy, Bar, Rod, Shapes, Tube and Wire, Extruded and Structural Shapes, General Specification For
QQ-A-225	Aluminum Alloy, Bar, Rod, Wire or Special Shapes, Rolled Drawn or Cold Finished, General Specification For
QQ-A-250	Aluminum Alloy, Plate and Sheet, General Specification For
QQ-A-435	Aluminum Alloy Sheet, Painted, for Exterior Use
SS-C-150	Cement, Bituminous Plastic
TT-E-527	Enamel, Alkyd, Lusterless
WW-T-700	Tube Aluminum and Aluminum Alloy, Drawn, Seamless, General Specification For

Military

MIL-P-116	Preservation, Methods Of
MIL-T-704	Treatment and Painting of Material
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-S-4174	Steel Sheet and Strip, Flat, Low Carbon Aluminum Coated

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MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Requirements For
MIL-C-5541	Chemical Films and Chemical Film Materials For Aluminum and Aluminum Alloys
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture Sensitivity
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-C-18969	Calking Compounds, Metal and Wood Seam

STANDARDS

Federal

FED-STD-147	Tape, Pressure Sensitive and Gummed, Methods of Inspection, Sampling and Testing
FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Procedure for the Selection Of
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing Of
MS-33586	Metals, Definition of Dissimilar

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

A153	Specification for Zinc Coating (Hot Dip) on Iron and Steel Hardware
B209	Aluminum Alloy, Sheet and Plate
B210	Aluminum Alloy, Drawn Seamless Tubes
B211	Aluminum Alloy, Bars and Rods
B221	Aluminum Alloy, Extruded, Bars, Rods, Shapes and Tubes

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B241	Aluminum Alloy, Seamless Pipe and Seamless Extruded Pipe
B308	Aluminum Alloy, Standard Structural Shapes, Rolled and Extruded
B313	Aluminum Alloy, Round Welded Tubes
B316	Aluminum Alloy, Rivet and Cold Headed Wire and Rods
B429	Aluminum Alloy, Extruded Structural Pipe and Tube
E96	Methods of Tests for Water Vapor Transmission

Uniform Classification Committee (UCC)

Uniform Freight Classification Rules

(Applications for copies should be addressed to the committee at 202 Union Station, 516 W. Jackson Boulevard, Chicago, Illinois 60606.)

Aluminum Association Publications

Specifications for Aluminum Structures
Aluminum Standards and Data
Guide for Design of Aluminum Formed-Sheet Building Sheathing

(Applications for copies should be addressed to the association at 750 3rd Avenue, New York, N.Y. 10017.)

Metal Building Manufacturer's Association (MBMA)

Recommended Design Practices Manual

(Applications for copies should be addressed to the association at 2130 Kieth Building, Cleveland 15, Ohio.)

American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc (ASHRAE)

Handbook of Fundamentals

(Applications for copies should be addressed to the society at 345 E. 47th Street, New York 17, N.Y..)

Technical Society and Technical Association specifications and standards are generally available from libraries. They are also distributed among technical groups and using federal agencies.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

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3.2 Description, general. The general requirements, design, clear span, length, clear height (from finished floor to bottom of inside roof framing at haunche), door openings, number and type of doors, number and type of windows, and other specific details shall be as specified in the detailed specification.

3.2.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein, shall be selected in accordance with MIL-STD-143 except as provided in 3.2.2 and 3.2.3.

3.2.2 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.3 Standard parts. With the exception in 3.2.2, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials.

3.3.1 Materials. Material shall be as specified herein. Materials not definitely specified herein shall be selected by the contractor and shall be subjected to all requirements of this specification.

3.3.2 Aluminum.

3.3.2.1 Structural aluminum. Aluminum for structural and framing members shall be 6061-T6 aluminum alloy in accordance with QQ-A-200, QQ-A-225, QQ-A-250 or ASTM designations B209, B211, B221, B241 or B308. Other aluminum alloys may be used provided they are subjected by either the producer or the purchaser to anaylsis, tests and other controls to the extent and in the manner prescribed by ASTM and federal specifications. Allowable design stresses shall be in accordance with applicable aluminum association publications referenced herein. When pipe or tubing is used for structural members it shall be of aluminum alloy meeting the requirements of ASTM designations B210, B241, B313, B429 or WW-T-700.

3.3.2.2 Aluminum covering.

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a. Aluminum covering shall be of alloy 3003, 3004, Alclad 3003, Alclad 3004, 5005, 5052 or 5154 and shall have a minimum thickness of 0.024 inches. Temper shall be such as to give the material the required mechanical properties to resist the design loads.

b. Aluminum clad steel covering in accordance with MIL-S-4174, type II may be used in lieu of aluminum covering.

3.3.3 Fasteners.

3.3.3.1 Bolts, nuts and washers. Bolts, nuts and washers for structural connections shall be as follows:

a. Bolts shall be of alloy designation 2024-T4, 6061-T6 or 7075-T73 and in accordance with ASTM designation B316.

b. Nuts shall be of alloy designation 6061-T6 or 6262-T9 and in accordance with ASTM designations B211 or B316.

c. Washers shall be alclad 2024-T3 or T4 or other alloy in accordance with ASTM designation B209. Spring lock washers shall be of alloy 7075-T6.

3.3.3.2 Sheet metal fasteners.

a. Fasteners for securing aluminum sheets or panels to structural supports shall be self tapping screws of aluminum alloy in accordance with this specification or 305 stainless steel. The screw shall have a hex head with aluminum (minimum thickness .050 inches) and neoprene washers, the minimum diameter of which shall be 5/8".

b. When screws are used for fastening sheet metal to sheet metal they shall be of 305 stainless steel or of aluminum alloy complete with washers as described in sub paragraph a above.

3.3.4 Dissimilar materials. Dissimilar materials as defined by MS33586 shall not be used in intimate contact with each other without suitable protection against electrolytic corrosion.

3.3.5 Extra material. The contractor shall furnish not less than an additional 5 percent of bolts, nuts, washers and other fastenings, sealant material and one gallon of paint when painting of the structure is specified. This additional material shall be of the same type and character supplied for the assembly of the building in the field.

3.4 Design and construction.

3.4.1 Design.

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a. The design of the aluminum structural components of the building shall be in accordance with the Aluminum Association Specifications for Aluminum Structures. Wind load design criteria shall be in accordance with the Metal Building Manufacturers Association Recommended Design Practices Manual. Unless otherwise specified in the detailed specification the building(s) shall be designed for a 75 mile per hour wind load and a 30 pound per square foot snow load. The following load combinations shall be used.

(1) Dead load plus snow load.

(2) Dead load plus wind load.

(3) Dead load plus snow load plus one-half wind load.

(4) Dead load plus wind load plus one-half snow load.

Under combinations c and d, a 33 percent increase in allowable unit stress is permitted.

b. Aluminum siding, roofing and other covering shall be designed in accordance with Aluminum Association publications "Aluminum Standards and Data" and "Specifications for Aluminum Structures." Minimum thickness shall be as specified herein (see 3.3.2.2).

3.4.2 Construction. The building, including all parts, packaging and packing shall be in accordance with this specification. Parts bearing the same assembly mark shall be alike and interchangeable. The building shall be fabricated to assure ease of assembly, fit, interchangeability of like components, proper operation of doors, and disassembly for complete salvageability and relocation except for joint mastic, damaged fasteners and insulation. The suppliers regular commercial configuration for wall and roof covering, and ridge panel may be used provided that the sheets will sustain without permanent deformation the design loading specified (see 3.4.1). Flashing trim, and rib filler stops solely for the utilization of the sheets offered will be permitted. The entire building shall be so fabricated that no welding, riveting, special tools, or special equipment shall be necessary for field erection.

3.4.3 Welding. Shop welding shall be in accordance with the applicable portions of the Aluminum Association publication "Specifications for Aluminum Structures". No field welding shall be necessary for the erection of the building.

3.4.4 Tolerances. The supplier shall be responsible for providing tolerances in accordance with industry standards to insure proper fit, assembly and operation of doors and windows.

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3.4.5 Hardware. All bolts, nuts, screws, washers and other hardware not covered by 3.3.3 necessary for the erection of the building shall be furnished and shall be of the best commercial grade. This hardware shall be of aluminum alloy 6061-T6 or 6063-T6. Stainless steel, aluminized steel or steel galvanized in accordance with ASTM designation A153 may be used in lieu of aluminum alloys for hardware, provided the requirements of 3.3.4 are met.

3.4.6 Reliability. The building shall be constructed to withstand the stresses and environmental conditions to which it may be subjected during the intended period of use of at least 10 years after a maximum storage period of 2 years.

3.5 Insulation. When specified (see 6.2) the roof and walls shall be insulated as specified herein. The insulation may be panel, blanket or board type.

3.5.1 Panel insulation. A metal sandwich type of panel insulation consisting of an interior and an exterior facing of aluminum sheet as specified herein (see 3.3.2.2) with a core consisting of insulation material in accordance with HH-I-521 or HH-I-558 shall be used when specified. The core shall be bonded to each facing and the U-factor of the panel, air to air, shall not exceed 0.21 when determined for winter conditions (15 mph wind) in accordance with ASHRAE Handbook of Fundamentals.

3.5.2 Blanket insulation. Blanket insulation shall conform to HH-I-558, Form I, Type I, Class 6. The insulation shall be covered with a factory applied aluminum facing (4 mil thick nimimum) having a water vapor transmission rate of not more than 0.10 Perm. when tested in accordance with ASTM E96. The U-factor shall be in accordance with 3.5.1.

3.5.3 Rigid insulation. Block or board insulation, rigid or semi-rigid, shall be in accordance with HH-I-558, Form A, Class 1 or 2 and shall meet the U-factor requirements of 3.5.1.

3.5.4 Insulation retainers. Insulation retainers of a type, size and design necessary to hold the insulation in place and provide a neat appearance shall be provided. Metalic retainers shall be non-ferrous or have a non-ferrous coating. Non-metalic retaining members including adhesives shall have a fire resistance classification of not more than that permitted for the insulation (see 3.5.1).

3.6 Foundation and anchor bolts.

3.6.1 Foundation. The building foundation shall not be required under this specification.

3.6.2 Anchor bolts. Anchor bolts, nuts and washers necessary to anchor the building to the foundation shall be of structural steel galvanized in accordance with ASTM designation A153. The bolts shall be of sufficient size and length to resist the column reactions resulting from the specified

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load combinations (see 3.4.1). All anchor bolts shall be provided with a hook at the unthreaded end to provide a positive locking device. When the column base plate is set, the provisions of 3.3.4 shall be met.

3.7 Wall and roof sheets. Wall and roof sheets shall be of material as specified herein (see 3.3.2.2 and 3.4.2). Purlin and girt spacing shall be determined by allowable loads on the roof and wall sheets. Sheets shall be fabricated with length and shape predetermined and accurately formed. All roof and wall sheet end laps shall be a minimum of six inches except that end laps for siding having a depth of section of less than two inches shall be a minimum of four inches. All sheets shall be shop fabricated for their position on the building, assuring ease of field erection. Either the ridge connection of the roof sheets shall be properly curved and lapped so that the lap can be made away from the prevailing winds to assure weather-tightness or a properly fitted ridge flashing shall be supplied. Sheet connections at doors, windows, and ends of sheet runs shall be so designed that a minimum of flashing is required to assure proper weathertight connections.

3.7.1 Sealant. Sealant for making weathertight the side laps, eave panes, roof ridge ventilator, flashing and other panel joints shall be butyl, silicone polysulfide or other material meeting the requirements of MIL-C-18969. Optional sealant may be bituminous plastic cement in accordance with SS-C-153. Self-adhesive aluminum foil tape meeting the requirements of Fed-Std-147 may be used as a sealant.

3.8 Doors.

3.8.1 Truck and tractor doors. Sliding or overhead truck and tractor doors shall be of the number, size and location specified in the detailed specification. Materials shall be as specified herein (see 6.2).

3.8.2 Personnel doors. Personnel doors shall be of the number, size, and location specified in the detailed specification. Personnel doors shall be of the hollow metal type constructed of not less than 0.032 thick aluminum alloy material as specified herein. The doors shall be stiffened and sound deadened with a honeycomb core laminated with high strength neoprene adhesives. Personnel doors shall be finished in accordance with this specification (see 3.13). Each door unit shall be furnished complete with an extruded aluminum alloy threshold, weather stripping, full mortice prime painted hinges in accordance with FF-H-116, panic hardware and all other necessary hardware and parts for complete door installation. Door hardware shall be in accordance with FF-H-106.

3.9 Windows. Windows shall be of the number, location, size, type (fixed, project out, sliding, etc) and performance level specified in the detailed specification. Windows shall meet the requirements of the Architectural Aluminum Manufacturers Association.

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3.10 Louvers and framed openings. Louvers and framed openings shall be of the number, size, type and location specified in the detailed specification. Framed openings shall be fabricated of aluminum alloy as specified herein (see 3.3.2.1) and designed to receive the specified size louvers. Louvers shall be adjustable, ventilating type, fabricated or extruded aluminum alloy sheet metal as specified herein (see 3.3.2.2). Louver blades shall pivot on nylon or other non metallic bearings and shall be opened by a pull chain and closed by return spring when the chain is released. Each louver shall have a free flow area of 45% and shall provide a minimum of 60% weather protection when in the full open position. An aluminum insect screen of 14 - 18 mesh shall be provided and attached to the exterior of the louver assembly. The complete louver assembly shall be finished in accordance with 3.13.

3.11 Other building accessories. Other building accessories such as roof jacks, roof vents, skylights, etc. shall be of the number, size, type and location as specified in the detailed specification and shall be fabricated from materials as specified herein. Finish shall be in accordance with 3.13.

3.12 Interchangeability. All parts having the same manufacturers part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000, Class 2, shall govern changes in the manufacturer's part number.

3.13 Finishes shall be as specified herein or as specified in the detailed specification (see 6.2).

3.13.1 Finish for structural and bracing members. Structural and bracing members of the building shall require no finish in addition to the natural oxide finish of aluminum.

3.13.2 Finish for roofing and siding sheets, accessories, etc. Where painting of sheet aluminum is not required no additional finish is necessary.

3.13.3 Painted aluminum surfaces. When specified in the detailed specification all or portions of the building shall be painted. The portions to be painted shall be given a conversion coating of amorphous chromate, phosphate or oxide meeting the requirements of MIL-C-5541. The surface shall then be primed with a zinc chromate primer in accordance with MIL-P-8585 and painted with a lusterless alkyd enamel in accordance with TT-E-527. Color shall be as specified in the detailed specification and in accordance with FED-STD-595. Shop painting of aluminum sheets in accordance with QQ-A-435 is permitted.

3.14 Identification marking. The building shall be identified in accordance with MIL-STD-130.

3.15 Erection marks. The erection mark of each component, structural member or piece shall be stenciled on the item or on an attached metal tag in the case of rods or other narrow items. Markings shall be gothic type capitals and arabic numerals not less than one inch high. Stenciling shall conform to MIL-T-704.

3.16 Erection manual. Erection manuals shall be furnished in accordance with Form DD-1423.

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3.17 Aluminum fabrication. Aluminum used in the fabrication of the building shall be free from kinks and sharp bends. The straightening of materials shall be done neatly and accurately. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Bolt holes shall be accurately punched or drilled and shall have all burrs removed that would deleteriously affect the function or erection of the part.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that materials used are manufactured, examined and tested in accordance with referenced standards and specifications.

4.2 Classification of tests. Tests to be conducted on the building shall consist of preproduction and acceptance tests.

4.2.1 Preproduction tests. Preproduction tests shall consist of the following tests performed on one building.

4.2.1.1 Assembly test. One complete building consisting of all the items furnished under the contract shall be erected at the contractor's plant under the supervision of the procuring activity. This preproduction erection shall demonstrate the satisfactory overall compliance with the drawings and this specification. Any shortages or inconsistencies in the drawings or material lists shall be resolved in writing with the procuring activity prior to submission of the building for preproduction acceptance.

4.2.1.2 Environmental tests. The environmental tests specified in the detailed specification or 6.2 of this specification shall be performed in accordance with MIL-E-5272.

4.2.2 Acceptance testing. Acceptance testing shall consist of the following.

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4.2.2.1 Mechanical inspection of members and parts. All members and parts shall be given a mechanical inspection to determine the quality of material and workmanship and compliance with the requirements of the contract. Particular attention shall be given to the following points:

a. Fit
b. Identification and erection reference markings
c. Finishes
d. Length
e. Corrosion prevention
f. Welded joints
g. Screw thread assemblies

In addition, each building shall be subjected to any other test specified herein which the inspector considers necessary to determine compliance with the requirements of this document.

4.2.2.2 Completeness. Each building shall be checked with a check list for completeness prior to shipment.

4.2.2.3 Conformance. An accurate check shall be made to determine that the production building is identical to the preproduction building. No deviation shall be made in either design, workmanship, quality, or quantity of items supplied.

4.3 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the building shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation and packaging for building components and accessories shall be determined and accomplished in accordance with MIL-P-116 and MIL-STD-794. Interior containers shall be determined and utilized in accordance with MIL-STD-794. The contents of each package shall contain items and quantities which will facilitate efficient and expeditious placement at the erection site for timely depacking and erection in established steps of succession called for in the erection manual.

5.1.1.2 Paint. Paint shall be preserved in accordance with MIL-P-116, method IC-2.

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5.1.1.3 Fasteners. Bolts, nuts, washers, clips, screws, and similar items of like description shall be preserved in accordance with MIL-P-116, method IC-2, except where small quantities (2 pounds or less) are involved, then method IC-3 may be used.

5.1.1.4 Technical publications. The technical publications required for each building shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.2 Level C. The complete building shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Packing and exterior shipping containers shall conform to the requirements of MIL-STD-794, except that large bar (framing), sheet, plate and structural shape components shall be packed in accordance with MIL-STD-163. The final arrangement of the shipping containers shall be in accordance with MIL-STD-794 and lifts conforming to MIL-STD-163 shall be entirely suitable for identification, documentation and accountability of all packages belonging to one (1) building, when multiple quantities of like buildings are being processed and handled simultaneously through shipping and receiving ports.

5.2.2 Level B. The complete building shall be packed as specified for level A.

5.2.3 Level C. The complete building shall be packed in a manner to insure arrival at destination in satisfactory condition and which will be acceptable to carriers at lowest rates. Packing shall comply with UCC Uniform Classification Rules.

5.3 Marking. Interior packages, bundles, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The building(s) covered by this specification is(are) intended for general use at CONUS or overseas bases.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Level of preservation, packaging and packing required (see 5.1 and 5.2).
c. Insulation requirements (see 3.5).

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d. Door requirements (see 3.8).
e. Window requirements (see 3.9).
f. Finish requirements (see 3.13).
g. Erection manuals (see 3.16).
h. Environmental tests required (see 4.2.1.2).

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